

A Taxonomy of Human-Robot Teamwork Requirements

Abstract—Autonomous systems are increasingly deployed in safety- and mission-critical domains where humans and robots must operate as a team to complete complex tasks. Existing requirements for Human-Robot teamwork remain fragmented across disparate sources, with no unified framework that addresses complexities of collaborative human-robot tasks. We address this gap by presenting a taxonomy of Human-Robot teamwork requirements derived from analysis of standards, regulatory guidance and the literature, both academic and industrial. We extracted 361 requirements from 14 cross-domain sources. Through iterative classification and refinement, we develop a two-level hierarchical taxonomy comprising 6 high-level categories and 21 lower-level subcategories that distinguish information provision, relational control, decision support, safety mechanisms, performance monitoring, and foundational system capabilities. Our taxonomy addresses both conceptual clarity (defining what constitutes Human-Robot teamwork requirements) and practical applicability (supporting requirement elicitation in real systems). To assess the validity of this framework, we conduct an expert evaluation with several domain specialists. We further examine how requirements distribute across Human-Led, Robot-Led and Shared operational perspectives, revealing responsibility boundaries that shape safe collaboration. This work provides a structured foundation for requirements engineering practitioners to systematically elicit and specify Human-Robot teamwork capabilities in critical domains, enabling reliable integration of autonomous systems into cyber-physical teams.

I. INTRODUCTION

Autonomous systems are increasing in complexity and capability, enabling new possibilities for collaboration between humans and robots in critical settings. Human-Robot teams are increasingly being tasked with completing complex operations in a plethora of domains, including autonomous robotic systems that (1) reduce human exposure to hazardous environments (e.g. nuclear applications [13]), and (2) enable operations that are simply not possible for humans in isolation due to hazardous and hostile environments (e.g. space missions [11]). In these critical settings, humans must work alongside AI-driven robots to complete tasks that require complementary capabilities, shared decision-making, and coordinated action.

Requirements are central to software development in critical, highly-regulated domains. Yet, for Human-Robot teamwork, they remain fragmented across disparate standards, industry technical reports, and academic literature. No framework exists that specifically addresses the teamwork dimensions of Human-Robot collaboration. This hinders the systematic deployment of Human-Robot teams in critical areas such as aerospace, nuclear operations, and advanced manufacturing, where comprehensive requirements are mandatory.

This work aims to fill this gap by providing a comprehensive taxonomy and corpus of software requirements for Human-Robot teamwork. By organising distributed and disparate Human-Robot teamwork requirements into a coherent hierarchy, we provide a structured foundation for designing Human-Robot teamwork, supporting the integration of AI into diverse cyber-physical teams. This structure supports traceability and comparison across systems and team structures, providing requirements engineers with a reference framework against which specifications can be assessed for coverage and clarity.

Contributions. In this paper, we contribute:

- **A corpus of 361 requirements** for Human-Robot teams that were collected by analyzing current standards and literature sources spanning multiple sectors (Section III).
- **A two-level hierarchical taxonomy** of requirements for Human-Robot teamwork comprising 6 high-level and 21 low-level categories, each characterized by distinct teamwork patterns (Sections IV and V).
- **An expert evaluation** validates the taxonomy’s conceptual clarity and explores of Human-Led, Robot-Led and Shared operational perspectives (Sections VI and VII).

These contributions provide a solid basis for requirement elicitation for Human-Robot Teamwork in critical domains.

II. BACKGROUND AND RELATED WORK

Human-Robot teams collaborate through *supervisory control* (humans oversee and intervene), *shared decision-making* (combining strengths to achieve goals), and *independent collaboration* (performing distinct tasks while sharing information). Their growing use in critical domains demands well-defined, verifiable requirements (as reflected in standards like DO-178C [30]). While AI systems are increasingly integral to such collaboration, current requirements focus on technical correctness while neglecting the critical aspects of teamwork, posing significant challenges to reliability and safety.

A recent NASA technical standard [29] defines human-system interaction requirement classes spanning crew health, performance, training, and hardware/software design. We draw inspiration from [29], reusing its nomenclature where applicable in our taxonomy. In addition to refining some of the categories from [29] to more accurately reflect Human-Robot teamwork, we provide several new ones that are not present in [29]. We discuss this comparison in Section VI.

The Brahms language has been used to provide semi-formal descriptions of Human-Robot teamwork for space applications [3], [37]. Recent work provides mission requirement patterns for autonomous mobile robots [36] but it does not capture the

Venue	Type	Reference
Advances in Industrial and Manufacturing Engineering	Journal	[31]
Conference on Human Factors in Computing Systems	Conference Proceedings	[1]
Towards Autonomous Robotic Systems	Conference	[16]
IEEE Intelligent Systems	Journal	[25]
IEEE/AIAA Digital Avionics Systems Conference	Conference	[35]
International Journal of Robotics Research	Journal	[38]
IEEE Access	Journal	[5]
MITRE	Tech Report	[28]
International Requirements Engineering Conference	Conference	[24]
ISO	Technical Standard	[21], [20], [22]
Federal Aviation Administration (FAA)	Technical Standard	[12]
NASA	Technical Standard	[29]
Code of Federal Regulations (CFR)	Technical Standard	[4]
National Transportation Systems Center	Tech Report	[40]

TABLE I
SOURCES INCLUDED IN THE REQUIREMENT CORPUS

human. The absence of formal specification patterns limits the ability to systematically specify (and subsequently verify) interactions in Human-Robot teams. Explainability requirements and associated patterns have been studied in [7], [33], [34]. Our taxonomy captures several explainability concerns but has a much broader focus on Human-Robot teamwork. In this way, our taxonomy compliments existing literature.

Other related work formalizes properties related to Human-Robot interaction using the tock-CSP formalism with the RoboScene tool [18]. In this work, the authors focus on capturing assumptions for mathematical reasoning about Human-Robot interaction. They validate their approach on a search-and-rescue drone use case. In contrast to our work, which focuses on requirements for Human-Robot teams, [18] provides a UML-style notation for defining interactions between humans and robots. The conditions that they include in their models are likely instantiations of some of the categories of our taxonomy but examining this in detail is left as future work.

III. METHODOLOGY

In this paper, we seek to answer two research questions:

RQ1: What are suitable categories of Human-Robot teamwork requirements?

RQ2: To what extent can these requirement categories capture diverse patterns of Human-Robot teamwork?

For **RQ1**, we consider categories as *suitable* if they are coherent and well-defined, cover human-robot teamwork requirement types without excessive overlap, and are judged useful by domain experts for elicitation and specification tasks. We address **RQ1** by deriving a structured taxonomy of Human-Robot teamwork requirements from technical standards, regulatory guidance, and peer-reviewed literature (Table I). **RQ2** focuses on validating the taxonomy’s completeness and coverage across different patterns of human-robot teamwork through expert evaluation.

We follow a four-stage methodology: (1) literature and standards review to identify requirement sources, (2) requirement extraction and corpus assembly from identified sources, (3)

iterative taxonomy development, and (4) validation through expert evaluation and make appropriate modifications.

Search Strategy: First, we employed a hybrid, multi-stage source identification approach. Initial sources were identified through expert recommendations and targeted searches of organization websites, such as NASA, FAA, and ISO. These initial sources were then expanded using snowballing [39] by examining references within relevant technical standards, reports and publications. We also examined literature using the IEEE Xplore, ACM Digital Library, and Springer digital libraries. Our search strings included combinations of: core terms, e.g., “human-robot team”, “human-robot collaboration”, “human-robot interaction”, combined with “requirement”, “specification”, “standard”, “safety”. Regarding the temporal scope, we did not apply strict start/end date restrictions, as foundational standards remain relevant.

Inclusion Criteria: Sources were included if they met all of the following: 1) contained explicit requirements, design guidance or specification patterns; 2) addressed human-robot interaction in collaborative or team-based settings; 3) focused on operational, safety, and/or teamwork aspects; 4) provided sufficient detail to extract requirement statements.

Exclusion Criteria: Sources were excluded if they: 1) focused solely on algorithmic performance without human interaction context; 2) addressed single-agent autonomous systems without human-robot collaboration; 3) provided high-level conceptual frameworks without specific requirements; 4) focused solely on shared goals or joint intentions in classical multi-agent systems literature (e.g. [23]).

Source Selection Process: We began with 72 initial candidate sources. Following title and abstract screening, 42 documents were retained for full-text review. Application of the inclusion and exclusion criteria resulted in a final corpus of 14 sources used for requirement extraction listed in Table I.

Extraction Process: Two researchers independently reviewed each source document to identify requirement statements. The researchers compared their extractions and discussed discrepancies. For ambiguous requirement statements, a third researcher reviewed to determine if the requirement statement followed the inclusion/exclusion criteria. Initial agreement between the two primary extractors was 87%. Disagreements primarily concerned: 1) Boundary between teamwork and general Human-Robot Interaction (HRI); 2) whether statements were requirements versus contextual information. For the purpose of extraction, we distinguish teamwork requirements from general HRI requirements based on interdependent collaboration. We retained only requirements that explicitly support coordinated roles, shared responsibility, or distributed decision-making toward a common objective. All disagreements were resolved through discussion and third-party review, resulting in the final corpus of 361 requirements.

Taxonomy Development: Once we assembled our corpus of requirements from these sources, we examined and classified each one. We adopted NASA’s Human-System Standard [29] as our initial resource since: 1) it represents a validated classification used in safety-critical aerospace ap-

plications; 2) it provides detailed human-system interaction categories; 3) it has been developed through extensive expert consultation and real-world application. However, preliminary analysis revealed that this framework was insufficient for complete coverage of our corpus, necessitating extension and refinement. The refinement process either broadened or narrowed the scope of those in [29]. To enhance reliability, requirement extraction and categorization were conducted iteratively, with repeated refinement of category definitions to minimize overlap and ambiguity. Specifically, we broadened the scope when we did not find many requirements fitting into a specific (and usually narrow) category. We narrowed the scope when several requirements fit a predefined category but there were clear, notable differences between them.

Taxonomy Structure: The taxonomy derived from our corpus of requirements is split into a two-level hierarchy by design, with high-level categories capturing abstract teamwork concerns, while low-level categories represent more concrete and actionable requirement groupings. These levels follow a one-to-many hierarchy: each high-level category may encompass multiple low-level categories, while every low-level category maps exactly to one high-level category.

Validation Approach: To address **RQ2**, we employed two complementary validation methods: 1) coverage assessment, which checked whether all extracted requirements were successfully classified (Section VI-A); 2) expert evaluation with 5 domain experts (Section VI-B).

IV. TAXONOMY: HIGH-LEVEL CATEGORIES

Our categorization provides a high-level classification of requirements for human-robot teamwork, organized by functional area (see Fig. 1 and Table II). Each high-level category reflects a functional concern within teamwork and differs in terms of *what is being exchanged* (e.g., data, commands, or safety measures), the *purpose of the exchange* (e.g., situational awareness, control, or safety assurance), and the *type of interaction* (one-way, two-way, proactive, or reactive).

INFORMATION SHARING AND AWARENESS addresses state visibility without prescribing action; CONTROL AND COORDINATION concerns allocation and authority; COMMUNICATION AND DECISION SUPPORT addresses influence on choice; SAFETY AND RISK MITIGATION focuses on protection and containment; TEAM PERFORMANCE MONITORING concerns assessment of agent state; and TEAM CAPABILITIES AND AUTHORITY defines foundational properties. The first three categories primarily focus on *real-time teamwork and decision-making*, ensuring shared understanding and synchronized action, whereas SAFETY AND RISK MITIGATION and TEAM PERFORMANCE MONITORING are oriented toward *ensuring safe, reliable, and effective collaboration*. Additionally, categories such as INFORMATION SHARING AND AWARENESS and TEAM CAPABILITIES AND AUTHORITY are predominantly proactive, while SAFETY AND RISK MITIGATION and TEAM PERFORMANCE MONITORING define primarily reactive interactions. These distinctions ensure analytical separability while acknowledging operational interdependence - particularly between INFORMATION SHARING AND

AWARENESS, which provides state visibility without directing behavior, and COMMUNICATION AND DECISION SUPPORT, which actively shapes action.

For each category we provide a description of the (1) rationale; (2) type of exchanged information; (3) type of interaction and (4) key differentiator(s) between it and the other categories. Each category is decomposed into several subcategories, for each one of which we provide a description of the rationale as well as example requirements in Section V.

A. INFORMATION SHARING AND AWARENESS

Rationale: This category focuses on the availability and accessibility of critical data for both human and robotic teammates. It emphasizes maintaining situational awareness and ensuring that necessary information is provided to both parties to enable effective collaboration.

Type of information: Raw data and status timely updates between human(s) and robot(s).

Purpose: Ensuring that both human(s) and robot(s) have the necessary information to understand the system's state and operate effectively.

Type of interaction: Primarily one-way. For example, robot providing data to human, or vice versa.

Key Differentiator: This category is about passive awareness. It ensures that operators are informed but it does not consider direct decision-making or control actions.

Example Scenario: The robot provides telemetry, automation status, and mode changes to ensure that the human has the necessary information.

B. CONTROL AND COORDINATION

Rationale: This category encapsulates the ability of human operators to configure, control, and delineate responsibilities within the human-robot team. It ensures that the robot is used and performs in a way that aligns with human intent while maintaining safety and operational efficiency.

Type of information: Commands, permissions, and role delineation between human and robot.

Purpose: Enabling the human to define, adjust, and/or authorize the robot's automated/autonomous actions while ensuring that the system operates within the defined framework.

Type of interaction: Two-way direction where the human configures the robotic automation settings and the robot operates within those constraints.

Key Differentiator: Unlike the "Information Sharing and Awareness" category, this category is about actively shaping how the robot functions within the team.

Example Scenario: The human sets robotic automation levels, defines responsibility boundaries, and/or authorizes system restarts after failures.

C. COMMUNICATION AND DECISION SUPPORT

Rationale: Requirements in this category center on the *provision and clarity* of information. This category highlights the role of communication in facilitating decision-making and understanding between human and robot teammates. It includes

Category	Focus	Interaction Type	Key Differentiator
INFORMATION SHARING AND AWARENESS	Providing system status and data	Primarily one-way (status updates)	Ensures situational awareness but does not shape actions
CONTROL AND COORDINATION	Managing tasks and system behavior	Two-way negotiation (commands and permissions)	Ensures the human defines automation's role and actions
COMMUNICATION AND DECISION SUPPORT	Aiding decision making	Interactive (recommendations and alerts)	Provides guidance rather than just raw data
SAFETY AND RISK MITIGATION	Preventing/responding to hazards	Proactive and reactive (safety features, failure responses)	Focuses on preventing harm and handling failures
TEAM PERFORMANCE MONITORING	Assessing team effectiveness	Continuous assessment (monitoring behavior)	Tracks ongoing performance rather than providing static data
TEAM CAPABILITIES AND AUTHORITY	Defining system capabilities and ensuring human authority	Static system design and emergency control	Establishes fundamental system abilities and human control mechanisms

TABLE II
HIGH-LEVEL CATEGORIZATION OF REQUIREMENTS

requirements that are related to decision aids, interface clarity and notifications to support effective human-robot teamwork.

Type of Information: The exchanged information includes processed information, recommendations, and alerts from the robot to the human.

Purpose: Helping the human to make better, faster, and more informed decisions while maintaining situational awareness and trust in the robot.

Type of Interaction: Interactive guidance, i.e., robot provides recommendations that the human interprets and acts upon.

Key Differentiator: Unlike the “Information Sharing and Awareness” category, which provides raw data for passive awareness, this category captures information provision that actively shapes human decision-making. This includes both on-demand decision aids and proactively volunteered information whose nature compels or strongly directs a human response, distinguishing it from information that merely updates situational awareness.

Example Scenario: A decision aid provides suggested actions, explains its reasoning, and notifies the human when it lacks sufficient data to provide recommendations.

D. SAFETY AND RISK MITIGATION

Rationale: This category comprises requirements that ensure safe operation within human-robot collaborative environments. It includes preventative safety measures, emergency responses, and failure recovery mechanisms to protect both human operators and robotic systems.

Type of information: Safety mechanisms, failure responses, and environmental safeguards between the human, robot, and workspace.

Purpose: Ensuring that the human-robot team operates within a safe framework, minimizing the risks to both human and robotic agents.

Type of interaction: Proactive safety measures like workspace design. Reactive safety measures like failure recovery.

Key Differentiator: Unlike the “Control and Coordination” category, which focuses on who does what, this category ensures that no matter who is in control, safety is prioritized at all times.

Example Scenario: If the robot fails, it enters a safe mode and/or provides alerts to the human. Safety zones ensure that humans and robots operate without interference.

E. TEAM PERFORMANCE MONITORING

Rationale: This category covers the continuous assessment of both human and robot performance to maintain situational awareness and ensure optimal teamwork. It includes monitoring human behavior as well as robotic system performance.

Type of information: Observations of human and robot behavior, performance metrics, and real-time monitoring data.

Purpose: Ensuring that both human and robotic agents are performing optimally and adjusting operations based on observed performance.

Type of interaction: Continuous assessment where the robot monitors human behavior and vice versa.

Key Differentiator: While the “Information Sharing and Awareness” category provides static data, this category focuses on continuous tracking and evaluation of human and robot performance.

Example: The system detects operator fatigue or degraded robot performance and adjusts operations accordingly.

F. TEAM CAPABILITIES AND AUTHORITY

Rationale: This category defines the fundamental capabilities of both robotic and human teammates, ensuring that robots are designed with characteristics that support effective collaboration, while also guaranteeing that humans maintain ultimate authority over the system.

Type of information: System design considerations, human-focused requirements, and override mechanisms.

Purpose: Establishing the fundamental capabilities of both human and robot teammates while ensuring that the human retains ultimate authority.

Type of interaction: Static system design and emergency control. The robot has built-in capabilities, but the human retains override authority.

Key Differentiator: Unlike the “Control and Coordination” category, which manages real-time interactions, this category defines the fundamental capabilities and limits of both human and robot teammates from the outset. More precisely, Control and Coordination governs the dynamic allocation of authority at runtime, whereas Team Capabilities and Authority establishes the structural constraints and override boundaries within which such allocation occurs.

Example Scenario: The robot’s designed capabilities (e.g., sensor accuracy, movement speed) determine how well it

can collaborate. Meanwhile, the human retains the ability to override or shut down the robot at any time.

V. TAXONOMY: LOW-LEVEL CATEGORIES

We decompose each of the high-level categories into several sub-categories. The relationship between the high- and low-level categories is shown in Fig. 1.

A. INFORMATION SHARING AND AWARENESS

1) Robot Data Availability

Rationale: Data, such as telemetry, states, modes, etc., should be saved and made available to operators.

Example Requirement:

“Automated/robotic systems shall record and make available operational and performance data to both crew and ground support personnel.” [29]

2) System Status Visibility

Rationale: The human operator needs to maintain situational awareness to calibrate trust in the system and avoid errors, requiring visibility into system health and projected state to assess likely robot performance. Awareness of performance decrements or failures is necessary to enable timely intervention or task takeover.

Example Requirement:

“Automated/robotic systems shall provide the human operator with the following information: system state and projection of future state, including failure or decrements in performance (e.g., battery power versus traverse distance and assessment of uncertainty in projection of future state).” [29]

3) Robot Mode Change Notification

Rationale: Clear indication of the current mode helps prevent mode errors, where operators take inappropriate actions or fail to act due to misunderstanding the state of the system. Notification by displays or other means gives the operator the ability to prepare for a mode change and/or to adjust behaviour to a new mode environment.

Example Requirement:

“The automated/robotic system shall notify the human operator of mode changes of any safety-critical operations.” [29]

4) Human Status Visibility

Rationale: As well as passing information from robot to humans (e.g. *System Status Visibility*), it is crucial that there is a way of transferring information from human to robot. This subcategory encapsulates the requirements relating to robots that must interpret and receiving information from humans.

Example Requirement:

“The automation/autonomy shall enable users to input information priorities and changes in those priorities.” [28]

B. CONTROL AND COORDINATION

1) System Initiation

Rationale: The human operator is to remain in control and must authorize a system restart after an emergency stop.

Example Requirement:

“If a detected failure occurs, autonomous operation only resumes after a deliberate restart from outside the collaborative workspace.” [20]

2) Robot System Configuration

Rationale: In a human-robot team, the human operator must remain in control and thus must be able to modify automation configuration parameters, including setup inputs, initial conditions, and termination conditions. However, certain configurations must remain restricted due to system-specific performance or safety constraints.

Example Requirement:

“Automated/robotic systems shall provide the human operator the ability to modify system configuration within the safety and performance limits of the system.” [29]

3) Responsibility Delineation

Rationale: A clear, salient delineation of the responsibilities of the human and the robot system is important to ensure that the operator performs the appropriate actions at the appropriate time. Transitions between robotic and human operator control need to be performed under well-defined procedures that specify the system state before and after the transition. Specific indicators will help to support human-robot interaction that avoids errors of omission and commission.

Example Requirement:

“Automated/robotic systems shall indicate whether a human operator or system is expected to perform a particular operation at a specific time.” [12]

4) Multi Agent Coordination

Rationale: Requirements specifically relating to the interaction between more than one agent in a team, e.g., the connection between and sharing of information amongst systems. This includes multi-robot and multi-human coordination.

Example Requirement:

“The automated/robotic system shall be able to make connections between requests from different agents, both human and automated.” [28]

5) Range of Control (Levels of Automation)

Rationale: Robotic systems are expected to be operated in different manners; for example, these highly flexible systems will be controlled directly (i.e., manual control) or commanded remotely (i.e., supervisory control). The multiple options associated with these highly flexible systems must be operable via a range of controls that are available to the human operator.

Example Requirement:

“Adaptive automation should be implemented at the point at which the user ignores a critical amount of information.” [12]

C. COMMUNICATION AND DECISION SUPPORT

1) Decision Support

Rationale: Decision aids provide pertinent information, analysis, and/or suggested solutions for continued operations. The system ultimately needs to enable the operator to make those decisions, whether or not it is the operator that acts on them.

Example Requirement:

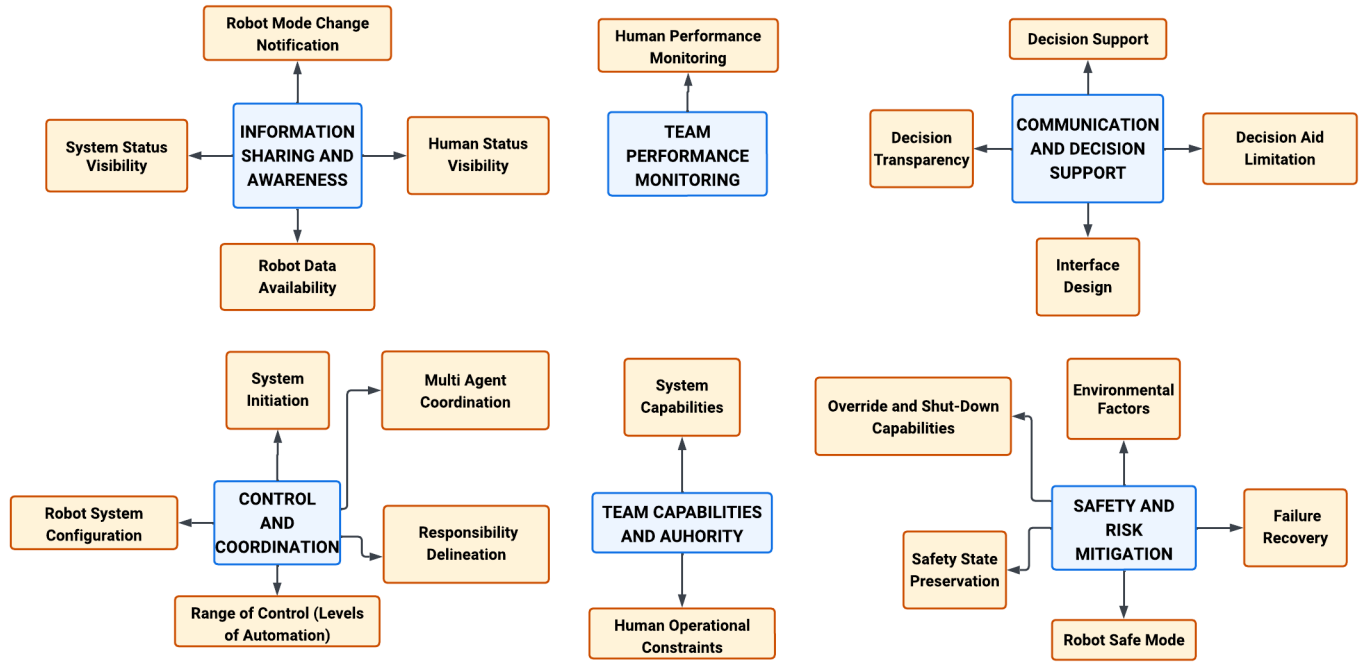


Fig. 1. Hierarchy of categories in our taxonomy. The high-level categories are shown in the blue boxes and the low-level categories are in the orange boxes.

“The automated/robotic system shall provide enough information about the developing event so that the operator can intervene effectively.” [28]

2) Decision Transparency

Rationale: The human operator needs to understand why the automated system is recommending actions, and the consequences of those actions, to make an informed decision without significantly disrupting the operator’s task, maintaining situational awareness, as well as calibrating trust.

Example Requirement:

“Decision aid systems shall provide explanations and rationales, and consequences of potential actions.” [29]

3) Decision Aid Limitation

Rationale: The human operator needs to be made aware when a decision aid is unable to assist with a decision due to lack of information or limitations in design so that information can be provided, or other avenues can be explored in a timely manner.

Example Requirement:

“Decision aids shall notify the human operator when a problem or situation is beyond the aid’s capability.” [29]

4) Interface Design

Rationale: Communication through displays and interfaces, including how and when warnings should be presented.

Example Requirement:

“Warning and caution alerts must provide timely attention-getting cues through at least two different senses by a combination of aural, visual, or tactile indications.” [4]

D. SAFETY AND RISK MITIGATION

1) Environmental Factors

Rationale: Collaborative workspaces must prioritize operator safety and efficiency. Environment considerations such as clearly defined collaborative zones, perimeter safeguarding, and protective measures ensure that robots operate safely within shared spaces, allowing for controlled entry and human oversight. These safeguards create a structured and adaptive environment where humans and robots work together securely.

Example Requirement:

“The design of the collaborative workspace shall be such that the operator can easily perform all tasks and the location of equipment and machinery shall not introduce additional hazards.” [21]

2) Safety State Preservation

Rationale: In a failed transfer of control from the automated system to the human, the automated system that is experiencing degraded performance must take appropriate measures to remain in a safe state, including alerting the operator of the degraded state and transfer of control. Other safety-related requirements also fit into this category.

Example Requirement:

“Automated/robotic systems not selected for operation remain in a safe state in a multi-robot cell.” [20]

3) Robot Safe Mode

Rationale: Protective actions include avoidance manoeuvres, protective stops, and “return home” if the human operator can no longer communicate or command the system. The transition to operator control needs to occur safely and minimize harm to the operator and vehicle.

Example Requirement:

“The automated/robotic system shall take protective action (e.g., avoidance manoeuvre, protective stop) or request that the operator safely take control if the system’s operational safety threshold is exceeded.” [29]

4) Failure Recovery

Rationale: If a failure or inability to function happens, the information provided by the system is to enable the human operator to collect confirming or exclusion evidence to determine a safe course of action.

Example Requirement:

“Early warning notification of pending automation failure or performance decrements should use estimates of the time needed for the user to adjust to task load changes due to automation failure.” [12]

5) Override and Shutdown Capabilities

Rationale: The system is to allow the human operator the ability to override or shut down robotic systems if it is determined that these systems present a risk, or if redirection of activities is needed. The human is to always remain in ultimate control of the vehicle throughout a mission. It is essential that the override or shut down capability is performed safely, i.e., avoids inadvertent harm to crew and vehicle.

Example Requirement:

“The automated/robotic system shall provide the operator the capability to override the automation and assume partial or full manual control of the system to achieve operational goal states.” [28]

E. TEAM PERFORMANCE MONITORING

1) Human Performance Monitoring

Rationale: The autonomous/robotic agent in the team must be able to monitor aspects of human behavior, performance and abilities to maintain understanding of the situation and tasks.

Example Requirement:

“The autonomous teammate shall monitor the actions and performance of the human teammate(s).” [35]

F. TEAM CAPABILITIES AND AUTHORITY

1) System Capabilities

Rationale: Robot capabilities and characteristics are critical in human-robot teamwork to determine the system’s ability to operate reliably, predictably, and collaboratively with human teammates. These requirements represent capabilities and characteristics that are essential for collaboration in teams.

Example Requirement:

“Automation should be designed to adapt by providing the most help during times of highest user workload, and somewhat less help during times of lowest workload.” [12]

2) Human Operational Constraints

Rationale: Requirements that focus on the human counterpart rather than the robotic system. This includes requirements relating to the actions and behaviors that a human should withhold, as well as system requirements that specifically aid the human teammate.

Example Requirement:

“Users should be given an adequate understanding (mental model) of how the automated system works in order to monitor effectively.” [12]

VI. METRICS AND EXPERT EVALUATION

The full list of requirements that we collected and the slides used to interview the experts are available in our repository [2].

A. Category and Sub-Category Metrics

Table III collates the number of requirements in our corpus of 361 requirements that fall into each category (Section IV) and sub-category (Section V) from our taxonomy. The majority of requirements (100/361) fall into COMMUNICATION AND DECISION SUPPORT, predominantly in the *Decision Support* sub-category.

The next largest category was TEAM CAPABILITIES AND AUTHORITY, primarily comprising the *System Capabilities* sub-category. Many of these requirements addressed handover tasks between humans and robots, as described in [38].

Several requirements in SAFETY AND RISK MITIGATION concerned safety mechanisms for robots operating in close proximity to humans, including provisions specific to handover tasks. In *Environmental Factors*, we found several requirements that referred to the design of the collaborative workspace and physical protective measures. These requirements are also relevant in multi-robot teaming situations.

Within CONTROL AND COORDINATION, many requirements fell into the *Range of Control (Levels of Automation)* sub-category. Many of these were concerned with the robot moving through different degrees of automation/autonomous functionality. This ranges from tele-operated robots through to those making decisions with little to no human intervention.

The largest sub-category within INFORMATION SHARING AND AWARENESS was *System Status Visibility*. This is unsurprising as the human must be aware of the current state of the robot at all times. This helps the human to make informed decisions based on their knowledge and trust in the robotic system. Several requirements also specified that the robot maintain awareness of the human teammate’s status, which we classified as *Human Status Visibility*.

In addition, we found 9 requirements concerned with *Human Performance Monitoring*. We did not find any related to monitoring robot performance but these likely exist in more complex scenarios than those available in the literature.

Table III indicates (shaded rows) which categories from our taxonomy have a corresponding category in [29]. As one of the most comprehensive existing standards in this space, [29] provided an important foundation, contributing 14 requirement categories and multiple requirement examples to our taxonomy. However, we have critically refined several of these categories, updating terminology based on evidence from our literature review and expert evaluation. For example, our *System Status Visibility* subcategory corresponds to *Automation System Status Provision* in [29]. We replaced ‘Automation’ with ‘Robot’ across several categories to better

reflect the scope of requirements identified in the literature, which extend beyond automation to encompass broader system components. We adopt ‘Visibility’ in place of ‘Provision’ to more accurately capture the intent of these requirements and to address terminology that consistently caused confusion during expert review. Crucially, our taxonomy extends beyond [29] by introducing 7 new categories derived from the wider literature. These categories have no counterpart in existing standards and represent a substantive original contribution to the systematic characterisation of human-robot teamwork requirements.

B. Expert Evaluation

To validate the taxonomy, we interviewed 5 experts with backgrounds in Human-Robot Teamwork, Human Factors and Requirements Engineering (2 academics, 2 from large robotics engineering companies and 1 from a space agency). Topics of discussion centred around validating the proposed taxonomy with respect to both its conceptual validity (**RQ1**) and practical applicability (**RQ2**). Participants were guided through each of the six high-level categories and asked questions focused on conceptual clarity and distinctness, classification rationale of sub-categories, as well as whether the respective category captured the intended class of human-robot teamwork requirements, aiming to directly address **RQ1**. To address **RQ2**, participants were asked to assess coverage and potential gaps by relating categories to real human-robot teaming systems they were familiar with and reflect on the usefulness of the taxonomy for requirements elicitation and specification analysis. We thematically analyze [26] the discussions into 5 topics:

1) Awareness vs Action: Experts recognized the separation between awareness and action as a useful operational distinction (INFORMATION SHARING AND AWARENESS, and COMMUNICATION AND DECISION SUPPORT, respectively). However, they noted that information initially intended to support passive understanding can, depending on the context, become relevant or urgent to decisions. As such, systems may need mechanisms to signal changing criticality while still preserving the conceptual boundary between awareness/informing and acting/directing.

Several experts emphasized the importance of shared Situational Awareness (SA) [10] relating to team members’ (human and robot) internal models of the world around them [8]. SA is important to ensure that both humans and robots maintain a shared and accurate understanding of the environment, tasks, and each other’s states [29]. This enables safe coordination, decision-making, and effective responses to unexpected changes. Rather than treating SA as a stand-alone construct, we can decompose it into operational requirement classes [9]. INFORMATION SHARING AND AWARENESS supports level 1 (perception) through visibility of system state, COMMUNICATION AND DECISION SUPPORT supports level 2 (comprehension) by providing structured rationale and interpretations, and TEAM PERFORMANCE MONITORING partially supports aspects of level 3 (projection) by anticipating performance decrements. However, projection in Human-Robot teams extends beyond these decrements and requires *anticipation* of authority transition

and shifts in responsibility and control, which is not fully represented in our taxonomy. Requirements that enable structured degradation and explicit authority delineation therefore operationalize Level 3 SA as a coordination property rather than merely a cognitive one, suggesting that SA in Human-Robot teams is closely connected with authority management.

2) Reduced Human to Robot Information and Negotiated Control:

While the taxonomy and extracted requirements provide structure for information from robot to human, there is less explicit description of what information the robot should know about its human teammate. Experts highlighted the importance of representing human priorities, condition, and behavior to enable better adaptation and coordination. Furthermore, experts challenged the implicit assumption that control flows primarily from human to robot, noting situations in which the robot may need to guide, influence, or temporarily supersede human action, particularly during error, uncertainty, or time pressure. These observations reframe control as a dynamic, bidirectional exchange grounded in reciprocal awareness rather than a fixed allocation of authority.

3) Ergonomics and Human Variability: Safety cannot be reduced to collision avoidance or fail-safe mechanisms; workspace design, human variability, and fatigue considerations must also be incorporated within ergonomic requirements. Experts further emphasized that effective teamwork requirements must accommodate variability in human behavior and attention, rather than relying on assumptions of consistent compliance. Together, these observations highlight the need to account for physical, cognitive, and behavioral variability in the design of safe and effective human-robot collaboration.

4) Human Performance Monitoring Challenges: All experts regarded this category as essential but challenging. Measuring physical performance is feasible but assessing cognitive state, intention, or understanding is more difficult. One expert extended this further by suggesting future systems may need explainability of human behavior as well as humans needing explainability from robotic behavior.

5) Additional Categories: Within TEAM PERFORMANCE MONITORING, several experts suggested further refinement into sub-categories such as *Human Performance*, *Robot Performance*, *Quality of Interaction*, and *Monitoring of Operational Parameters*. These suggestions reflect a desire to distinguish between physical performance metrics, cognitive or behavioral assessment, and interaction-level dynamics. In our taxonomy, these dimensions are consolidated within a single monitoring category to maintain structural coherence. However, in future refinements, we will consider distinguishing these aspects where empirical evidence indicates divergence in their functional implications.

Overall, experts regarded the taxonomy as conceptually coherent and practically useful for requirement elicitation, while also identifying areas for refinement, particularly in modelling human cognition.

Category	Total	Human-Led	Robot-Led	Shared
INFORMATION SHARING AND AWARENESS	56	N	Y	Y
<i>Robot Data Availability</i>	6	-	Y	-
<i>System Status Visibility</i>	35	-	Y	-
<i>Robot Mode Change Notification</i>	10	-	Y	-
<i>Human Status Visibility</i>	5	Y	-	Y
CONTROL AND COORDINATION	59	Y	N	Y
<i>System Initiation</i>	4	Y	-	-
<i>Robot System Configuration</i>	14	Y	-	-
<i>Responsibility Delineation</i>	13	Y	Y	Y
<i>Multi-Agent Coordination</i>	5	Y	Y	Y
<i>Range of Control (Levels of Automation)</i>	23	Y	-	-
COMMUNICATION AND DECISION SUPPORT	100	Y	Y	Y
<i>Decision Support</i>	49	Y	Y	Y
<i>Decision Transparency</i>	23	-	Y	-
<i>Decision Aid Limitation</i>	3	-	Y	-
<i>Interface Design</i>	25	Y	Y	Y
SAFETY AND RISK MITIGATION	61	Y	Y	Y
<i>Environmental Factors</i>	11	Y	Y	Y
<i>Safety State Preservation</i>	10	Y	Y	Y
<i>Robot Safe Mode</i>	18	-	Y	-
<i>Failure Recovery</i>	11	Y	Y	Y
<i>Override and Shut-Down Capabilities</i>	11	Y	-	Y
TEAM PERFORMANCE MONITORING	9	-	Y	-
<i>Human Performance Monitoring</i>	9	-	Y	-
TEAM CAPABILITIES AND AUTHORITY	76	Y	Y	Y
<i>System Capabilities</i>	64	Y	Y	-
<i>Human Operational Constraints</i>	12	Y	-	-

TABLE III

THE NUMBER OF REQUIREMENTS IN OUR CORPUS PER CATEGORY AND SUB-CATEGORY. WE CLASSIFY THE CATEGORIES AS HUMAN-LED, ROBOT-LED OR SHARED. THE TOTAL REQUIREMENTS FOR AN INDIVIDUAL CATEGORY IS THE SUM OF THOSE FOR ITS SUB-CATEGORIES. THOSE COLORED IN A GRAY BACKGROUND HAVE AN ANALOGOUS CATEGORY IN THE “AUTOMATED AND ROBOTIC SYSTEMS” PART OF [29].

VII. DISCUSSION

A. Human-Led, Robot-Led and Shared Classification

To answer **RQ2**, we analyze the classification based on three operational perspectives: (1) **Human-Led**, (2) **Robot-Led** and (3) **Shared** (Table III) to expose patterns of authority and interdependence often implicit in existing standards. Robot-Led functions emphasize autonomous stabilization and information provision; Human-Led functions preserve oversight and override authority; Shared functions regulate coordinated interaction. This framing clarifies how authority boundaries shape safe collaboration beyond simple task allocation.

1) *Structural Boundaries of Authority*: Our ternary classification aligns with established models of autonomy such as Sheridan’s 10 Levels of Automation (LoA) [32]. Specifically, we map Human-Led to Low LoA (1–4), Robot-Led to High LoA (7–10), and Shared to Mid LoA (5–6). This operational analysis reinforces the distinction between runtime authority allocation (CONTROL AND COORDINATION) and structural authority constraints (TEAM CAPABILITIES AND AUTHORITY).

Robot-Led safety requirements include autonomously entering *safe mode* in response to dangerous conditions and executing *failure recovery* by detecting and reporting system failures. While failure detection is Robot-Led, the choice of response is typically Human-Led, and the resultant execution is shared for coordinated failure handling.

For effective handover, the robot must communicate error severity and type, not just generic failures. Therefore, the *Failure Recovery* sub-category exists for the robot to convey information that allows the human to identify and classify the progression of failure that triggers a well-informed response.

Human-Led subcategories include the authority to override or shutdown the robot in unsafe situations (*Override and Shut-Down Capabilities*) which is essential to meet legal and ethical requirements, especially in regulated domains. Most subcategories within CONTROL AND COORDINATION are Human-Led, reflecting the need for retained human supervisory authority. This override authority interacts with the Robot-Led safety functions in SAFETY AND RISK MITIGATION (e.g., *Robot Safe Mode*). While the robot can autonomously stabilize or enter a safe state, human override remains the final control layer when automated safeguards fail. This supports prior work on shared control in physical human–robot interaction where dynamic arbitration of control authority through intent detection and feedback mechanisms [27] is emphasized. These authority boundaries become especially critical during failure, when control must transition under time and safety constraints.

2) *Graceful Degradation as a Teamwork Property*: A particularly critical concern in Robot-Led teamwork is graceful degradation [19], i.e., the ability of the system to fail safely and support a structured, predictable transition of control back to the human in the event of a failure or performance decrement. As the degree of robot autonomy increases, the human operator becomes progressively removed from active control, making the quality of this transition increasingly consequential. A system that degrades gracefully will alert the human with sufficient lead time, maintain a safe state during the transition, and ensure the human can effectively assume control. Requirements addressing graceful degradation are captured primarily within SAFETY AND RISK MITIGATION subcategories, which together define the mechanisms by which a Robot-Led system can fail without catastrophic consequences.

Safety is a shared function in areas requiring risk management and coordinated failure handling. Shared safety protocols require both human and robotic agents to perform coordinated risk management activities. Effective safety degradation requires explicit specification of how authority transitions during failure, who stabilizes, who diagnoses, and who determines subsequent action. Shared safety protocols and structured responsibility shifts ensure that control transfer occurs predictably and without introducing additional risk, whether that be during Human-Led, Robot-Led or collaborative tasks.

While degradation concerns how control transitions under failure, effective collaboration also depends on how well teammates model each other during normal operation.

3) *Shared Awareness*: Beyond authority transitions, effective teamwork requires reciprocated modelling between agents. The lack of requirements concerning human cognitive state, adaptability and reciprocal awareness suggests that existing standards prioritize system functionality over modelling the human as an active teammate. For example, if the robot detects elevated cognitive strain in the human, (within *Human Perfor-*

mance Monitoring), it may adjust its communication frequency or physical behavior accordingly. Expert feedback reinforced this, highlighting both the limited representation of human cognitive modelling in current requirements and the difficulty of specifying observable indicators of intention, understanding, or trust. Recent work [6] explores user indicators that may prompt explanations, suggesting a pathway toward specifying observable measures of mutual understanding.

A related challenge is vigilance decrement [15], [17], i.e., in supervisory roles, prolonged low workload or monotonous activity reduces alertness and increases susceptibility to error. Requirements for human performance monitoring must therefore account for both overload and under-stimulation. More broadly, COMMUNICATION AND DECISION SUPPORT is inherently shared, i.e., the robot recommends and informs while the human evaluates and authorizes, and its effectiveness depends on the human’s ability to understand the rationale behind robot recommendations. These cognitive, authority, and coordination dynamics are deeply interconnected and must be treated as such in requirements specification.

4) *Interdependence*: Shared functions regulate interdependence by coordinating communication, role delineation and risk management, to support efficient and effective collaboration and teamwork. These requirements manage the complexities of interdependence, demanding that both (human and robot) agents engage in mutual communication and coordination to mitigate shared risk and ensure role compatibility. For example, CONTROL AND COORDINATION is shared when role delineation and transition procedures are in place, and INFORMATION SHARING AND AWARENESS is shared when status updates are bidirectional, i.e., from robots to humans but also from humans to robots. When functions are shared between human and robotic agents, clear delineation of authority is essential. Ambiguity over who is ultimately in charge during shared tasks is a known contributor to coordination failures and safety incidents [14]. Even in collaborative arrangements where responsibilities are distributed, requirements should explicitly establish which agent holds final authority.

Many of the categories and sub-categories are *Shared* (Table III), which underscores the need for a comprehensive delineation of responsibilities in human-robot teamwork scenarios. This undoubtedly extends to teams that involve multiple human and robot agents working to achieve a shared goal. These would be explicitly considered in the *Multi-Agent Coordination* and *Responsibility Delineation* categories.

Together, these observations indicate that teamwork requirements are structured by authority flow, degradation management, and reciprocal coordination rather than by hardware configuration or the number or type of agents involved.

B. Threats to Validity

We have endeavoured to be comprehensive throughout the development of our taxonomy but recognise several threats to validity and limitations of this work. To begin, our corpus was obtained via a structured literature review (Section III). Any literature review is subject to *coverage error*, meaning

that we may have missed relevant literature that would have been revealed if we had used different search terms. We tried to mitigate this by starting our search with well-established literature in the field and applying both forward and backward snowballing, in addition to systematic keyword searches. *Source selection bias* may have also been present since we focused on English-language sources and Western standards.

While the expert evaluation serves as a key mechanism for mitigating coverage-related threats and refining the taxonomy, it is itself subject to limitations. Our panel of experts, though deliberately recruited from diverse backgrounds spanning academia, industry, and a space agency, may not be fully representative of the breadth of human-robot teamwork expertise. Nevertheless, convergent feedback between all experts provides confidence in the taxonomy’s broader applicability.

There is a wealth of standards in the aerospace domain, since it is global and highly-regulated, a dominance of aerospace-related results poses in a threat to *domain coverage*. However, rather than undermining the taxonomy, this reflects the current state of the field, i.e., aerospace and industrial robotics represent some of the most mature and documented areas of human-robot teaming, making them an appropriate foundation for an initial taxonomy. We explicitly recognise that emerging domains may surface additional requirement types, and a cross-domain validation study is planned to extend and stress-test the taxonomy accordingly.

Finally, the process of categorizing requirements into a taxonomy involves interpretive judgement. To mitigate *construct validity* threats, the taxonomy was developed iteratively across multiple researchers and refined through expert review, ensuring that classification decisions were not made unilaterally. Crucially, all 361 requirements were successfully classified without the need for a residual “other” category, which is an indicator of structural completeness with respect to the current corpus. We acknowledge that requirements from under-represented domains may challenge category boundaries in the future, and we welcome such challenges as evidence of the taxonomy’s growing utility and reach.

VIII. CONCLUSIONS AND FUTURE WORK

This paper presents a two-level taxonomy of Human-Robot teamwork requirements derived from 361 requirements across standards and literature sources. The taxonomy provides a structured framework for organizing interaction concerns that are otherwise dispersed across domain-specific guidance. In practice, the taxonomy can support elicitation, review, and traceability by ensuring that authority boundaries, monitoring mechanisms, and coordination structures are explicitly specified alongside functional capabilities.

Future work will focus on two priorities: first, broader validation across diverse human-robot team configurations, including large-scale industrial case studies; and second, an investigation into the extent to which taxonomy categories can be formalized to support verifiable requirement patterns.

Data Availability: Our corpus is publicly available in [2].

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